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Counter-Cyclical Market Price Efficiency and Stock Liquidity

Abstract

This study establishes a comprehensive empirical framework for constructing counter-cyclical market efficiency factors. The study investigates the cyclical relationship between economic changes and changes in stock price efficiency, and their impact on stock liquidity as well as the characteristics of financial market mechanisms. The study finds that for most periods in China, counter-cyclical phenomena exist, meaning that economic downturns (negative changes in China's GDP) is negatively correlated with increases in market efficiency. Furthermore, counter-cyclical factors significantly elevate commonality in liquidity while reducing the liquidity of securities. Finally, a joint examination of the interaction between commonality in liquidity and counter-cyclical factors reveals that this interaction further diminishes stock liquidity. Again, under the interaction of counter-cyclical factors, the financial market liquidity-improving mechanism works by lowering interest rates, reducing the accuracy of information, decreasing market trading correlations, and increasing stock volatility, in order to enhance the liquidity of securities. Finally, this paper finds that the government guarantee effect (with state-owned membership as a proxy variable) reduces the impact of trading comovement on stock liquidity under the influence of counter-cyclical factors on market efficiency.

Keywords: counter-cyclical; market efficiency; commonality in liquidity; China State-owned firms

Declaration of generative AI and AI-assisted technologies in the writing process

During the preparation of this work the author(s) used ChatGPT 4 in order to improve the conciseness and clarity of our manuscript writing. After using this tool/service, the author(s) reviewed and edited the content as needed and take(s) full responsibility for the content of the publication.

1. Introduction

The financial crisis that began in 2007 exposed a fundamental question in asset pricing: how does the business cycle affect market efficiency? More precisely, does an economic downturn increase or decrease the informational efficiency of stock prices? If market efficiency varies systematically with the cycle, what are the implications for liquidity, price discovery, and the functioning of market microstructure?

Capital markets in developed countries are assumed as operating with minimal government interference. Governments provide and enforce the legal framework for property rights and contracts, regulate market activity, and, at times, intervene to stabilize markets during periods of distress. Through monetary and fiscal policies—particularly interest rate adjustments and liquidity provision—governments influence asset valuations, corporate financing conditions, and risk-taking behavior. In extreme circumstances, governments may also intervene directly through bailouts or emergency measures to prevent systemic collapse.

The extent and nature of intervention differ across institutional environments. China's financial system is characterized by frequent and direct government involvement, whereas developed countries policymakers typically limit substantial intervention to crisis periods. Under certain economic conditions, optimal policy may generate a government-centric equilibrium in which intervention is sufficiently pervasive that investors rationally devote resources to learning about policy disturbances rather than firm fundamentals. This framework, consistent with Brunnermeier, Sockin, and Xiong (2022), reflects a policy regime that prioritizes financial stability over other objectives.

In the literature on government intervention, general policy goals include improving informational efficiency, stabilizing prices, and maintaining market liquidity (Naranjo and Nimalendran, 2000; Pasquariello, 2007; Pasquariello, Roush, and Vega, 2020; Brunnermeier, Sockin, and Xiong, 2022). In such settings, the government can be viewed as trading against noise-induced price fluctuations to stabilize markets. By enhancing the informativeness of prices, these policies can improve allocative efficiency and, in turn, real investment decisions.

China provides a useful empirical setting to study these mechanisms. As the world's second-largest financial system, it operates under a distinct institutional structure while remaining exposed to global dollar liquidity conditions. The research framework builds on the logic of Holmström and Tirole (1998). In their setting, when firms face moral hazard and the economy is subject to aggregate liquidity or reinvestment shocks, the supply of liquidity for private sector is generally inefficient. Under such conditions, government policy that relaxes financing constraints or stabilizes liquidity can generate Pareto improvements by mitigating inefficient liquidation and underinvestment. Counter-cyclical policies operated by China government can increase the ability of enterprises to withstand economic downturns and shocks to financial market price efficiency—the information efficiency impact from counter-cyclical policies can activate the mechanisms of financial markets and change the covariance and the level of stock liquidity.

This study develops an empirical framework to measure and explain this cyclical variation adjustments prompted by China regulators. I construct a countercyclical market efficiency factor based on estimating the relation between macroeconomic conditions and stock price efficiency. Countercyclical efficiency is defined as an improvement in price efficiency during economic downturns (periods of negative GDP growth). I test this definition by regressing measures of price efficiency on economic activity, proxied by GDP growth, and then examine how the cyclical efficiency patterns affects stock liquidity and market mechanism.

Following Rajan and Zingales (1998), this study links macroeconomic fluctuations to firm-level outcomes through financial market frictions. Firms choose between short-term projects and long-term productivity-enhancing investments such as R&D. The completion of long-term projects depends on the firm's ability to withstand interim liquidity shocks. If financing is unavailable midway, these projects must be sold or liquidated.

Because of information asymmetry and liquidity distribution discrepancy across firms, liquidity is not efficiently allocated to the financial market. Firms facing mild liquidity shocks may hold excess funds, while firms facing severe shocks might experience shortages. As a result, some firms rely primarily on retained earnings to finance reinvestment, even when aggregate liquidity in the market is sufficient.

If financial market imperfections bind primarily during downturns, then managing the volatility of market efficiency over the economic cycle increases the probability that long-term projects survive during liquidity shocks. In this case, liquidity-induced financing constraints are relevant in recessions but largely irrelevant in expansions. Moreover, firms whose characteristics make them more exposed to liquidity risk benefit more from reductions in cyclical variations in market efficiency, as their projects become more likely to survive during adverse shocks.

The implication is straightforward to real economic activities. Counter-cyclical financial market policies that stabilize market information efficiency over the business cycle improve the survival of long-term investment projects and have positive real effects on listed firms in China. The main findings of this study are as follows. First, this study documents that, for the majority of sample periods in China, market efficiency exhibits a pronounced counter-cyclical pattern. Specifically, improvements in measured market efficiency are negatively correlated with changes in GDP growth, indicating that efficiency tends to increase during economic downturns and suggesting that China stock market information quality is countercyclical. This pattern contrasts with the conventional view that recessions impair price discovery. Instead, it suggests that adverse macroeconomic conditions intensify information production or that policy intervention reshapes the informational environment of markets. The evidence is consistent with an active role of government in managing market conditions during downturns.

Second, this study finds that the counter-cyclical efficiency factor has significant implications for liquidity dynamics. In particular, increases in the counter-cyclical factor are associated with higher commonality in liquidity, indicating that individual stock liquidity becomes more synchronized across the market. At the same time, the factor is linked to a decline in the overall level of stock liquidity. This combination implies that while liquidity conditions become more system-wide and coordinated, the depth of individual securities deteriorates.

Third, when this study examines the joint effect of counter-cyclical efficiency and liquidity commonality, the interaction term enters negatively and significantly. That is, periods experienced by both elevated counter-cyclical efficiency and high liquidity commonality experience a further reduction in stock liquidity. This result suggests that the amplification of liquidity co-movement under counter-cyclical regimes exacerbates liquidity contraction at the firm level.

Literature on government interventions in the financial system especially during crises, focuses primarily on the assistance or bailout of financial institutions (Bianchi, 2016; Bond and

Goldstein, 2015). In contrast, this study provides policy implications on commonality in liquidity related market mechanism that under the interaction of countercyclical factors, suggesting that the financial market mechanism conducted by policy guidance needs to lower interest rates, reduce earnings dispersion, decrease market trading correlations, and increase stock volatility to enhance stock liquidity.

Finally, Given the government's direct intervention/trading in the market, during crisis as well as during normal times, an important question is how to evaluate the performance of these interventions. In particular, this study finds that the government guarantee effect (with state-owned membership as a proxy variable) reduces the impact of trading commonality on liquidity under countercyclical market efficiency factors. These findings are consistent with literature that when the government-based economic outcome arises, investors all focus on learning about noise in future government intervention, and their trading, consequently, exposes the asset price to anticipated government noise in the future, rather than the fundamental (Brunnermeier et al., 2022).

This study first contributes to the literature analyzing the cyclicality of information production in business cycle by documenting that market efficiency in China varies systematically over the business cycle and that this variation has first-order implications for stock liquidity (Asriyan et al., 2022 ; Cooper and Weitzner,2026). Cooper and Weitzner (2026)provide evidence that bank loan information quality improves as local economic conditions deteriorate, , suggesting that banks' information quality is countercyclical. Following Fama (1970) and Kyle (1985), this study defines that market efficiency as the extent to which prices reflect firm-specific information and measure price informativeness using stock return synchronicity (R^2) as in Morck, Yeung, and Yu (2000) and Durnev et al. (2003). Extending this framework dynamically, this study shows that during economic downturns—measured by negative changes in GDP growth—price informativeness increases rather than deteriorates.

This finding challenges the conventional presumption that recessions impair price discovery. Instead, by inversing the macroeconomic conditions, China government intensifies investors' incentives to acquire and trade on information in China's highly regulated market. By showing that informational efficiency itself is counter-cyclical, the study complements the literature on economic state-dependent information production (Morck et al., 2000; Durnev et al., 2003; Chordia, Roll, and Subrahmanyam, 2008) and provides a new empirical dimension to the efficient market hypothesis that efficiency is not constant but varies predictably with macroeconomic conditions.

Importantly, this study shows that higher price informativeness in downturns might worsen adverse selection and raises commonality in liquidity, leading to significant reductions in stock liquidity. This establishes a direct empirical link between counter-cyclical efficiency and liquidity fragility, a connection largely absents in the existing literature.

Second, the paper contributes to the literature on market mechanisms by documenting how financial markets accommodate shocks when counter-cyclical efficiency factors interact with commonality in liquidity. This study shows that when price informativeness becomes high in downturns, trading comovement intensifies and liquidity deteriorates. Under these conditions, accommodating mechanism may function through lower interest rates, reduced information precision, reducing transaction correlations, and higher volatility.

These patterns suggest that when informed trading becomes too dominant and adverse selection threatens market participation, institutional forces effectively introduce noise to restore trading activity and prevent liquidity dry-ups. This mechanism is consistent with equilibrium interpretations of China's financial system management (Brunnermeier, Sockin, and Xiong, 2022), where stability is achieved not by maximizing price informativeness but by managing the balance between information production and participation.

This evidence provides rare empirical support for the idea that markets regulate their own information environment in response to shifts in efficiency. It also complements studies of government trading in currency, bond, and equity markets (Naranjo and Nimalendran, 2000; Pasquariello, 2007; Pasquariello et. al , 2020; Barbon and Gianinazzi, 2019) and theoretical work on how intervention affects information aggregation (Bond and Goldstein, 2015; Goldstein and Huang, 2016; Goldstein and Yang, 2019; Cong, Grenadier, and Hu, 2020). This study shows that these effects operate through the endogenous relaxing earnings dispersions and trading behavior, not only through direct liquidity provision.

Third, and most importantly, the study contributes to the literature on government intervention by showing how China institutional structure shapes the equilibrium relation between information and liquidity. Using state-owned enterprises (SOEs) as a proxy for implicit government guarantees, the study finds that government backing firms significantly weakens the negative impact of trading comovement on liquidity under counter-cyclical efficiency regimes.

This result demonstrates that government intervention affects not only market stability but also the process of information aggregation in asset prices.

China’s institutional background makes this mechanism observable. Since the late 1990s, preferential credit allocation and implicit guarantees to state-supported sectors have created asymmetric environments between SOEs and private firms. Investment dominates GDP, household income is relatively low, and local governments provide implicit financial insurance to SOEs (Shyu, 2017). Geng and Pan (2024) study China's credit market using a structural default model that integrates credit risk, liquidity, and bailout. They document improved price discovery and a deepening divide between state-owned enterprises (SOEs) and non-SOEs. Amidst liquidity deterioration, the presence of government bailout helps alleviate the heightened liquidity-driven default, making SOE bonds more valuable and widening the SOE premium. Meanwhile, the increased importance of government support makes SOEs more sensitive to bailout, while the heightened default risk increases non-SOEs' sensitivity to credit quality. Examining the real impact, this paper finds severe performance deteriorations of non-SOEs relative to SOEs, reversing the long-standing trend of non-SOEs outperforming SOEs. These features generate systematic differences in how information shocks transmit to liquidity across firms.

More broadly, this evidence complements the literature on counter-cyclical policy. This study shows that market efficiency is a potential policy manipulating market channel. By influencing the cost and accuracy of information processing, government intervention can regulate trading behavior and mitigate liquidity crises without relying on credit expansion.

Taken together, these contributions show that counter-cyclical efficiency, liquidity dynamics, and government guarantees are jointly determined in equilibrium. Government intervention does not simply stabilize markets, rather it reshapes the information environment in which markets operate, revealing a fundamental tradeoff between financial stability and price efficiency.

2. Literature review

The Efficient Market Hypothesis (Fama, 1970) classifies market efficiency according to the speed with which prices incorporate information. In its semi-strong form, prices reflect all publicly available information, while in the strong form they incorporate both public and private information. Subsequent research in market microstructure refines this framework by emphasizing not merely the speed of adjustment but the extent to which prices embed private, firm-specific information. Kyle (1985) demonstrates that even in a semi-strong efficient market, the degree to which prices reflect informed traders' private signals can vary systematically with market depth and trading intensity. Accordingly, this study adopts price informativeness—defined as the extent to which stock prices reflect idiosyncratic firm-level information—as a proxy for market efficiency.

A large body of empirical research employs the market-model R^2 as a measure of stock return synchronicity and, by extension, price informativeness. Several studies interpret a low R^2 as evidence of inefficiency (Bramante et al., 2015). The earliest applications of the market-model R^2 as a measure of price inefficiency were conducted by Morck et al. (2000) at the national level and by Durnev et al. (2003) at the firm level. Morck et al. (2000) conceptualize R^2 as a measure of stock return synchronicity and document that synchronicity is higher in countries with weaker property rights protection. They argue that weak investor protection suppresses arbitrage activity and discourages firm-specific information production, causing stock prices to be driven more by macroeconomic and political factors than by firm fundamentals. This interpretation established R^2 as a widely used proxy for price inefficiency in settings characterized by institutional frictions.

Durnev et al. (2003) further examine how to interpret the R^2 statistic, specifically whether firm-specific volatility reflects noise or enhanced price efficiency. To address this question, they analyze the relationship between current stock prices and future earnings. Their results show that firms and industries with lower R-squared values exhibit higher price informativeness, in the sense that current prices contain more information about future performance. These findings support the view that lower synchronicity corresponds to greater informational efficiency. Together, Morck et al. (2000) and Durnev et al. (2003) provide the empirical and conceptual foundation for using R^2-based measures to evaluate the extent to which markets incorporate firm-level information.

Building on this framework, the present study examines how market information efficiency adjusts over the economic cycle. Arbitrageurs rely heavily on accounting information—such as earnings dispersion and profitability volatility—to evaluate firm performance and implement trading strategies, including short selling. The effectiveness of these strategies depends on both the availability of reliable information and the cost of processing it. During periods of market stress, heightened uncertainty and liquidity constraints may increase information-processing costs and reduce arbitrage capacity. By influencing these costs, policy interventions may alter trading behavior and potentially mitigate liquidity dry-ups during financial crises.

China's institutional structure provides a particularly relevant context for examining these mechanisms. Investment accounts for a substantial share of Gross Domestic Product (GDP), while household consumption and labor income remain comparatively low. Since 1996, following the adoption of industrial restructuring policies that emphasized the development of heavy industry, credit allocation has been systematically biased toward state-supported sectors. In particular, local governments have provided implicit guarantees for long-term bank loans to state-owned enterprises (SOEs), creating an operating environment distinct from that of private firms. These institutional features generate asymmetric financing conditions and heterogeneous expectations of government support.

Given this structure, state-owned firms are widely expected to receive implicit support during periods of economic distress, or at least to benefit from greater investor confidence during system-wide liquidity shortages. Political sensitivity, preferential access to resources, and differences in governance arrangements may further distinguish SOEs from private firms.

Consequently, trading behavior, liquidity dynamics, and the informational content of prices may differ systematically between state-owned and non-state-owned enterprises, particularly during crises.

China also differs from Western economies in the frequency and scope of government intervention in financial markets. Whereas policymakers in advanced economies typically avoid direct market intervention outside periods of acute crisis, the Chinese government intervenes more actively to stabilize markets and guide expectations. Under certain conditions, optimal policy may generate a government-centered equilibrium in which investors place greater weight on policy signals than on firm fundamentals (Brunnermeier et al., 2022). This framework implies that liquidity contagion mechanisms may differ between China and the United States, leading to systematic cross-country differences in price informativeness and liquidity comovement.

Most research on countercyclical stabilization focuses on monetary policy. The People's Bank of China (PBoC) emphasizes maintaining adequate liquidity, stabilizing prices, optimizing policy transmission, reducing financing costs, and ensuring exchange rate stability. Its primary tools include interest rate adjustments, reserve requirement ratios, and open market operations. The stated policy objectives underscore the importance of aligning money supply growth and social financing expansion with macroeconomic conditions while preserving financial stability.

Empirical evidence on the effects of accommodative monetary policy is mixed. Guo et al. (2022) find that expansionary policy does not primarily drive leverage growth but instead stimulates aggregate demand more strongly than debt accumulation. Their firm-level analysis suggests that debt-to-revenue ratios provide a more accurate measure of micro-level leverage dynamics than aggregate indicators. In contrast, Jin et al. (2014) examine bank risk-taking behavior and show that loose monetary policy relaxes credit standards and induces excessive risk-taking. They recommend the use of countercyclical capital buffers to offset these unintended consequences.

Beyond conventional monetary policy, government intervention in financial markets encompasses a broader range of instruments, including regulatory adjustments, supervisory guidance, guarantees, bailouts, and targeted taxes or subsidies. These interventions operate not only through direct asset purchases but also through financial intermediaries that transmit policy effects to prices, liquidity, and information aggregation. As a result, government participation can influence multiple dimensions of market quality, including price volatility, trading liquidity, and informational efficiency.

A growing literature investigates how direct government trading affects market outcomes. Naranjo and Nimalendran (2000) and Pasquariello (2007) analyze currency market interventions and demonstrate that government trading can alter both liquidity and informational efficiency under specific policy objectives. Naranjo and Nimalendran (2000) decompose intervention into expected and unexpected components and find that the variance of unexpected intervention is associated with a statistically and economically significant increase in bid-ask spreads, whereas expected intervention has no significant effect. Pasquariello, Roush, and Vega (2020) examine Federal Reserve trading in the U.S. Treasury market and show that official participation influences price formation and trading conditions even in highly developed markets. These findings suggest that government trading can meaningfully affect the process by which information is incorporated into asset prices.

Intervention is even more frequent in emerging markets. Brunnermeier, Sockin, and Xiong (2022) study government intervention in the Chinese stock market and show that trading against noisy retail investors can reduce price volatility while simultaneously affecting informational efficiency. Huang, Qiu, Wang, and Wang (2022) analyze a related setting in which the government trades strategically based on private information, thereby shaping price discovery. These studies highlight how government participation can alter incentives for private information acquisition and change the informational content of prices.

Empirical evidence from advanced economies further illustrates the effects of large-scale asset purchases. Grosse-Rueschkamp, Steffen, and Streitz (2019), Todorov (2020), and Corell, Mota, and Papoutsi (2024) examine the European Central Bank's corporate bond purchase program and document its impact on asset prices, liquidity, and market functioning. Similarly, Barbon and Gianinazzi (2019), Charoenwong, Morck, and Wiwattanakantang (2021), and Li (2022) study the Bank of Japan's equity purchase program and show that direct government participation influences stock price dynamics and trading conditions. While these studies provide important empirical evidence regarding the consequences of intervention, they also underscore the need for theoretical and empirical frameworks that explain how such policies affect information aggregation and liquidity in equity markets.

A related theoretical literature examines government intervention during financial crises, particularly in the context of bailouts of financial institutions(Bianchi (2016), Bond and Goldstein (2015), Chari and Kehoe (2016), Lucas (2019), Philippon and Schnabl (2013)). Shyu (2017) analyze how expectations of government assistance influence stock liquidity during downturns. Bond and Goldstein (2015) show that uncertainty about future intervention can distort information aggregation in prices when intervention affects firms' real outcomes. Cong, Grenadier, and Hu (2020) study intervention in money market mutual funds within a global games framework, emphasizing how policy alters the public information available to investors facing coordination problems. Angeletos, Hellwig, and Pavan (2006) and Goldstein and Huang (2016) examine how an informed policymaker can design actions that transmit information and coordinate private responses, whereas Goldstein and Yang (2019) show that public disclosure by policymakers may reduce price informativeness by weakening incentives for private information acquisition.

This study considers government intervention in a broader institutional context and focuses specifically on its implications for stock liquidity and liquidity commonality across firms. If investors anticipate that government backed firms are more likely to receive government support during periods of systemic stress, their trading responses to market-wide liquidity shocks may differ across firms. Such heterogeneity can alter patterns of liquidity comovement and affect the cyclical behavior of price informativeness.

These issues are particularly salient in China, where SOEs are widely perceived to benefit from implicit government backing. Shyu (2017) argues that SOEs effectively receive financial insurance through expected bailouts, as investors anticipate government intervention to prevent failures that could threaten social stability. When SOEs encounter financial distress, investors often expect the government—acting as the ultimate controlling shareholder—to provide support. In contrast, private firms lack comparable protection and must bear the full impact of financing constraints during crises. This asymmetry implies systematic differences in investor expectations, trading behavior, and liquidity dynamics between state-owned and non-state-owned firms.

The global financial cycle provides an additional layer of complexity. Liquidity contractions often occur simultaneously across markets, reflecting synchronized movements in asset prices, leverage, credit growth, and capital flows. Monetary policy plays a central role in driving these dynamics. Following U.S. monetary tightening, global financial intermediaries deleverage, risk aversion increases, credit spreads widen, and capital flows contract (Miranda-Agrippino & Rey, 2015). Because the U.S. dollar serves as the primary funding currency for global banks, changes in U.S. policy affect their financing costs and leverage decisions. Through discount rate channels and carry trade incentives, monetary policy influences asset pricing worldwide.

Empirical evidence from Gertler and Karadi (2015) and Bekaert et al. (2013) documents the transmission of monetary policy through its effects on risk premia, interest rate spreads, and volatility. These mechanisms raise corporate financing costs and amplify financial conditions domestically. Expectations of declining economic activity and shifts in discount rates are rapidly incorporated into stock prices, often leading to abrupt and significant market declines.

Most existing research on countercyclical measures focuses on the effects of monetary policy (Pasquariello, 2007）. Pan Gongsheng, Governor of the People's Bank of China, proposed, 'Effectively implement a moderately loose monetary policy. Use a comprehensive array of monetary policy tools to maintain ample liquidity, ensuring that the growth of social financing scale and money supply aligns with economic growth and overall price level expectations. Consider promoting a reasonable rebound in prices as an important factor in guiding monetary policy, and work to keep prices at a reasonable level. Improve the transmission mechanism of monetary policy, better manage the relationship between stock and incremental resources, focus on revitalizing stock financial resources, and increase the efficiency of fund usage. Strengthen the guidance of central bank policy rates, enhance the implementation of interest rate policies, and promote reductions in financing costs for enterprises and lending costs for residents. Fully utilize both the overall and structural functions of monetary policy tools and continue to effectively advance the financial 'five major initiatives. Adhere to a market supply and demand-based approach, adjust with reference to a basket of currencies under a managed floating exchange rate system, fully leverage the market's decisive role in exchange rate formation, stabilize market expectations, and maintain the RMB exchange rate at a basically stable and reasonably balanced level.

First, maintain reasonable growth in financing and the overall money supply. Strengthen counter-cyclical adjustments, and adjust and optimize policy intensity and pace appropriately based on domestic and international economic and financial conditions and the operation of financial markets. Closely monitor changes in the monetary policies of major overseas central banks, continuously strengthen analysis and monitoring of liquidity supply and demand in the banking system and changes in financial markets, and conduct open market operations flexibly and effectively. Make comprehensive use of various monetary policy tools such as interest rates and reserve requirement ratios to maintain ample liquidity. While preventing funds from being idle or misused, guide financial institutions to deeply explore effective credit demand, maintain reasonable growth in monetary credit, and ensure that the growth of total social financing and the money supply aligns with economic growth and overall price level expectations.

To sum up, most countercyclical studies focus on monetary policy. The PBoC emphasizes maintaining adequate liquidity, stabilizing prices, optimizing policy transmission, reducing financing costs, and ensuring exchange rate stability. Tools include interest rates, reserve requirements, and open market operations.

Empirical studies show mixed effects. Literature finds that accommodative policy does not primarily drive leverage growth but stimulates demand more strongly than debt accumulation. Firm-level analysis shows debt-to-revenue ratios better capture micro leverage dynamics. Jin et al. (2014) study bank risk-taking and show that loose policy relaxes credit standards, inducing excessive risk. They recommend countercyclical capital buffers to offset this effect.

3. Model Specification

3.1 Model

This paper first analyzes the formation mechanisms that affect counter-cyclical mechanisms, thereby providing empirical data support for the impact of counter-cyclical mechanisms on market efficiency and corporate liquidity. This paper adopts the empirical design method of Rajan and Zingales (1998), which combines macroeconomic data to study the micro-level effects on firms.

(1) Counter-cyclical factor estimation

The cyclical market efficiency (β_{cyclical}) cannot be directly observed, nor is there a direct measure for it. Therefore, this study uses first-stage regression to estimate the cyclical changes in market efficiency for each month in the sample. This study defines the estimated cyclical market efficiency as regressing the change in market efficiency on changes in GDP, with the model specified as follows:

$$\triangle\ Infeff = \alpha + \beta_{\text{cyclical}} \triangle\ GDP + \varepsilon. \qquad (1)$$

△GDP reflects the position in the economic cycle during the period; α is a constant. Equation (1) is estimated separately for each period in the sample using rolling 24 months data for β_{cyclical}. A positive regression coefficient (β_{cyclical}) (and negative vice versa) indicates counter-cyclical market efficiency (and pro-cyclical vice versa), meaning that the average synchronicity of individual stock returns with the market return increases (△Infeff positive change) during economic upturns (△GDP increasing). An increase in average market synchronicity represents a decline in the overall market efficiency. In this study, similar to the methodology of Kamara, Lou, and Sadka (2008), the trend of monthly changes in overall market efficiency (△Infeff) is calculated as:

$$\Delta lnfeff_t = \log\left(lnfeff_t / lnfeff_{t-1}\right) \quad (2)$$

$lnfeff_t$ is the monthly average market price efficiency in month t.

(2) Model of the Effect of Counter-cyclical Factors on Stock Liquidity

In order to examine the impact of market efficiency counter-cyclical factors on liquidity, this study regresses the liquidity of securities in month t on the interaction terms between financial market mechanism factors and market efficiency counter-cyclical factors. The model is set as follows:

$$Amihud = \alpha + \beta_1 cyclical + \beta_2 Z + \beta_3 cyclical * Z + Controls + \varepsilon \quad (3)$$

Amihud's monthly average illiquidity ratio, calculated using daily trading data (Amihud, 2002), represents the daily absolute return per unit of trading volume, with a higher ratio indicating lower liquidity. Z is a factor representing the financial market's influence mechanism on market liquidity, including: commonality in liquidity (Liqrsq), market interest rates, earnings dispersion, etc. According to the hypothesis, this paper expects: $\beta 3 > 0$, indicating that the counter-cyclical market efficiency factor affects individual stock liquidity through financial market mechanisms.

3.2 Variable definition

（1） Amihud illiquidity ratio（Amihud）

Specifically, Amihud's illiquidity ratio is calculated using daily transaction data, which represents the absolute value of the daily return per unit of transaction value. A higher Amihud 's illiquidity ratio represents lower liquidity, and daily stock illiquidity is calculated as

$$Illiquidity = \frac{|r_t|}{dvol} \quad (4)$$

dvol is the dollar volume of transactions on day t, and $|r_t|$ is the absolute daily stock return. This ratio indicates the price change caused by the transaction amount per unit.

（2） Measurement of market prices efficient (Infeff)

The Efficient Market Hypothesis (EMH) posits that market prices fully reflect all available information and has been widely applied in theoretical models and empirical studies of financial security prices, providing fundamental insights into the price discovery process. Market efficiency reflects the speed and completeness with which the market absorbs public and private information. Therefore, this paper uses information efficiency as a proxy variable for analyzing market efficiency.

Roll (1988) pointed out that the degree of co-movement in stock prices depends on the relative amount of firm-level and market-level information' reflected in stock prices. Therefore, a higher degree of co-movement indicates lower information efficiency. Based on this, this paper uses the R-square of a regression model of individual stock returns on day t-1 market returns, day t market returns and day t+1 market returns as a proxy variable (*Infeff*) for market efficiency analysis. Using daily data, the paper calculates the R-square for each stock in month t, and then averages the R-square values of all stocks in the sample for month t to represent the overall market efficiency for that month.

（3） Nominal GDP（NomGDP）

Chang et al. (2015) used seasonally adjusted quarterly nominal GDP (NomGDP) value added and interpolated it with seasonally adjusted monthly nominal retail sales of consumer goods, nominal exports, nominal imports, and nominal industrial value added.

（4） Real GDP（RealGDP）

The monthly series by Chang et al. (2015) was constructed in two steps. In the first step, the monthly series was obtained by dividing nominal GDP by the monthly GDP deflator obtained in the first step. In the second step, Chang et al. (2015) performed interpolation to adjust monthly real GDP to match the total quarterly real GDP. Nominal consumption: they seasonally adjusted the monthly series of retail sales of consumer goods.

（5） Commonality in liquidity（Liqrsq）

In order to examine the relationship between the counter-cyclical market efficiency mechanism and commonality in liquidity (Liqrsq), this study follows the single-factor commonality in liquidity research methods of Hameed et al. (2010) and Karolyi et al. (2012), using the R-square of individual securities' liquidity to the market liquidity regression model as a proxy variable for the commonality in liquidity factor (Liqrsq). The model is specified as follows:

$$Amihud_{i,t} = \alpha + \beta_1 Amihud_{m,t} + \epsilon_{i,t} \qquad (5)$$

（6） State-owned enterprise（State）

This study defines the nature of the ultimate controller in the CSMAR database as a state-owned enterprise if it belongs to a state-owned enterprise or state-owned institution (State, 1 indicates a state-owned enterprise, otherwise 0).

（7） Interest rate（Shibor）

Since interest rates reflect the cost of funds in the credit market, the higher the interest rate, the higher the cost of capital for investors, which affects their ability to provide for corporate innovation. The Shanghai Interbank Offered Rate (Shibor) is the arithmetic average of the RMB interbank lending rates quoted by a panel of banks with higher credit ratings, and it fully reflects changes in market supply and demand for funds, serving as a benchmark interest rate in the money market. Therefore, this paper uses the Shanghai Interbank Offered Rate (Shibor) to analyze the impact of interest rate factors on the supply side of funds.

（**8**） Earnings dispersion (SDE)

This study follows Dichev and Tang (2009), using the standard deviation of earnings over the past eight quarters as a proxy for earnings dispersion to measure earnings dispersion (SDE), where a higher value indicates lower precision.

（**9**） Correlated Trading (TRD)

This paper follows the research method of Karolyi et al. (2012), using the R-square of the regression model of individual security turnover rate on market turnover rate as a proxy variable for trading correlation (TRD).

（**10**） Controls variable

Other control variables (Controls) include: market factors that affect corporate innovation and susceptibility to commonality in liquidity: the logarithm of total market capitalization of A-shares (Size), stock returns (RET), stock return volatility (RET_std), and price-to-book ratio (PB).

3.3 Data

The data used in this study comes from the CSMAR database, mainly including trading data and financial data of listed companies. U.S. data comes from CRSP trading data and Compustat financial data. Therefore, this paper selects monthly data from the CSMAR database for the years 2010 to 2016. Next, the paper excludes samples with missing trading or financial data, and all continuous variables are winsorized at the 1% level. This paper uses the GDP data comes from Federal Reserve Bank of Atlanta China's Macroeconomy: Time Series Data, which is regularly updated. A major advantage of this database is that the data of annual, quarterly, and monthly frequencies are updated semiannually.

4. Empirical Results

4.1 Descriptive statistics

Table 1 presents the descriptive statistics of the research data. From Table 1, this study finds that during this research period, the average value of commonality in liquidity (Liqrsq) was 0.2840, and the median was 0.2333, which is higher than the international average level of 0.2 (Karolyi et al., 2012), indicating that commonality in liquidity in China is relatively high. Furthermore, the average value of the countercyclical factor (cyclical) was 1.9732, and the median was 2.1088. The positive values imply that in more than half of the periods, China experienced countercyclical market efficiency, that is, during periods of economic upturn (positive △GDP changes), market synchronicity increased (positive △infeff changes). Increased market synchronicity represents a decrease in overall market efficiency.

4.2 Regression analysis

Table 2 The impact of market efficiency counter-cyclical factors and commonality in liquidity on securities liquidity. Columns 1 and 2 examine the relationship between commonality in liquidity and market efficiency counter-cyclical factors. This study finds that the economic counter-cyclical factor significantly increases the risk measure of stock liquidity, proxied by commonality in liquidity.

Columns (3) and (4) of Table 2 examine the joint effect of commonality in liquidity and the countercyclical market efficiency factor on individual stock liquidity. The coefficient on commonality in liquidity (Liqrsq) is significantly positive, indicating that higher aggregate liquidity risk is associated with low stock-level liquidity. This finding is consistent with Pastor and Stambaugh (2003) and Karolyi et al. (2012), who document that market-wide liquidity shocks propagate to firm level.

The coefficient on the countercyclical market efficiency factor (cyclical) is also significantly positive. This result implies that when market efficiency increases during economic downturns manifested by lower return synchronicity and higher incorporation of firm-specific information, inducing stock liquidity declines.

More importantly, the interaction terms between commonality in liquidity and the countercyclical efficiency factor (*cyclical_real* × *Liqrsq and cyclical_nominal* × *Liqrsq*) are both significantly positive. This result indicates that the adverse effect of commonality in liquidity on stock liquidity is significantly amplified when countercyclical market efficiency factor (cyclical) increase.

Due to frictions in the financial market, arbitrage has its limitations. The impact of a single factor on the financial market often requires considering not only its direct effects but also the results and pathways of its interactions with other factors. For example, changes in market structure, such as rising interest rates, can increase traders' funding costs, potentially forcing arbitrageurs to sell the securities they hold at lower prices, thereby affecting liquidity.

Therefore, this study further examines the impact of financial market factors (such as stock volatility and market interest rates) on stock liquidity formation. Column 1 of Table 3 tests the effect of stock volatility (*Ret_std*) on stock liquidity. The regression results show that the coefficient of stock volatility (*Ret_std*) is significantly positive, indicating that higher individual stock volatility reduces stock liquidity. It also tests whether the counter-cyclical market efficiency factor strengthens the effect of stock volatility (*Ret_std*) on liquidity. The regression coefficient of the interaction term representing the counter-cyclical market efficiency factor and security volatility (*cyclical_real* * *Ret_std*) is significantly negative, indicating that, the counter-cyclical market efficiency factor interacted with the volatility of securities in China can increase stock liquidity.

Since interest rates reflect the cost of funds in the credit market, the higher the interest rate, the higher the cost of capital for investors, which imposes a systematic constraint on their ability to provide liquidity. Therefore, this paper also analyzes the impact of market interest rates on stock liquidity. Column 2 of Table 3 examines the effect of market interest rates (Shibor) on stock liquidity. The regression results show that the coefficient representing market interest rates (Shibor) is significantly positive, indicating that an increase in market interest rates will reduce stock liquidity. Column 2 of Table 3 also tests whether the counter-cyclical factor of market efficiency amplifies the impact of market interest rates (Shibor) on stock liquidity. The regression coefficient of the interaction term between the counter-cyclical market efficiency factor and the Shanghai Interbank Offered Rate (cyclical_real * Shibor) is significantly positive, indicating that after the counter-cyclical factor of market efficiency is applied, China's market interest rates have a incremental significant effect on stock liquidity.

This study then explores the impact of factors influencing investors' trading motives for individual securities (as opposed to a basket of securities in an index) on stock liquidity. It argues that earnings dispersion affects investors' motivation to trade specific securities. Since gathering information for stock investment is costly, if a target company carries earnings dispersion, the returns from trading such firms will be more uncertain, leading investors to prefer market-wide portfolios, which in turn reduces the liquidity of individual stocks.

This study predicts that earnings dispersion affects stock liquidity, because information risk reduces investors' accuracy in valuing target securities, thereby decreasing the motivation to invest in individual securities and shifting funds toward market-level investment targets, such as indices, which in turn increases the decline in market-level efficiency. This study uses the measure of earnings dispersion (SDE) from Dichev and Tang (2009). The regression results show that the regression coefficient representing earnings dispersion (SDE) is not significant, indicating that earnings dispersion (SDE) has no significant effect on liquidity. Next, this paper examines whether the countercyclical market efficiency factor strengthens the impact of earnings dispersion (SDE) on liquidity. The regression results in column 3 of Table 3 show that the regression coefficient of the interaction term representing the countercyclical market efficiency factor and earnings dispersion (cyclical_real*SDE) is significantly negative, providing evidence that the countercyclical market efficiency factor reduces the impact of earnings dispersion (SDE) on stock liquidity.

Finally, this paper analyzes the impact of the correlated trading behavior on liquidity. The regression results in column 4 of Table 3 show that the regression coefficient representing correlated trading is significantly positive, indicating that correlated trading has a significant impact on liquidity. Column 4 of Table 3 examines whether the countercyclical market efficiency factor strengthens the impact of correlated trading (TRD) on stock liquidity. The regression coefficient of the interaction term representing the countercyclical market efficiency factor and trading commonality (Cyclical_real * TRD) is significantly positive, providing evidence that the countercyclical market efficiency factor amplifies the impact of correlated trading (TRD) on stock liquidity. Overall, the countercyclical market efficiency factor enhances the influence of financial market mechanisms on liquidity.

4.3 Government Guarantee Effect: Transmission Channels of Market Efficiency Counter-Cyclical Factors on Liquidity Dry-Up between state-owned firms and non-state-owned firms

China's investment accounts for a relatively high proportion of its Gross Domestic Product (GDP), while the corresponding shares of consumption, labor income, and household income are relatively low. The strong positive correlation between investment and household consumption was interrupted in the late 1990s. These changes began in March 1996, when the 8th National People's Congress passed a historic long-term plan to adjust the industrial structure over the next 15 years to strengthen heavy industry. Preferential credit policies for heavy industry—where local governments provided implicit guarantees for long-term bank loans to the sector—crowded out short-term loans for small businesses.

Due to the particularity of China's institutional background, government intervention and control of state-owned listed companies is a common practice in China. The government tends to implement intervention and control through measures such as providing financing support to state-owned listed companies. Given that there are differences in resource acquisition between state-owned listed companies (State-owned) and privately-owned listed companies (Non-state-owned), the impact of market crises on state-owned enterprises and private enterprises may differ. Therefore, in Table 4, this study conducts separate tests for state-owned firms and non-state-owned firms. The first four columns present the results for the private sector samples, while the last four columns present the results for the state-owned samples.

Columns 1 and 4 of Table 4 examine the impact of financial market mechanisms that shape market efficiency and the countercyclical factors of market efficiency on liquidity, respectively.

The test results indicate that the impact of financial market mechanisms that shape market efficiency and countercyclical factors of market efficiency on liquidity does not differ significantly across state-owned firms and non-state-owned firms.

It is worth noting that in Column 8, the study examined whether the market efficiency countercyclical factor strengthens the impact of trading commonality (TRD) on systemic liquidity. The significance of the regression coefficient for the interaction term between the market efficiency countercyclical factor and correlated trading (cyclical_real * TRD) disappeared, revealing a government guarantee effect (with state-owned samples as a proxy variable), which reduces the impact of the interaction between the market efficiency countercyclical factor and correlated trading (TRD) on liquidity.

5. Cross-Market Contagion of Financial Liquidity Risk and the Cyclical Efficiency of Market Liquidity

The global financial cycle refers to the phenomenon that fluctuations in financial activities around the world are interconnected. Its characteristics include the co-movement of risk asset prices, financial intermediary leverage, credit growth, and total global capital flows. In particular, liquidity tends to decline globally. In addition to pessimistic expectations being transmitted through falling financial markets, the monetary policies adopted are also driving factors of the global financial cycle. After monetary tightening in the United States, global financial intermediaries will significantly deleverage, overall risk aversion will increase, global factors of asset prices and global credit will contract, corporate bond spreads will widen, and total capital flows will decrease.

China's economic model involves regular and intensive government intervention in financial markets, whereas Western policymakers typically avoid substantial intervention except during crises. Under certain potential economic conditions, optimal government policies can lead to a government-centric equilibrium, where intervention is so intensive that all investors choose to acquire private information about policy noise rather than fundamentals. This policy framework reflects the Chinese approach, which prioritizes financial stability over other policy objectives (Brunnermeier et al., 2022). Therefore, this paper separately calculates cross-market contagion of financial liquidity risk and the cyclical efficiency factor of market liquidity in China and the United States, anticipating differences between the two.

5.1 Model

This study first analyzes the formation mechanisms affecting market liquidity and market efficiency, thereby providing empirical support for the influence of financial market mechanisms on stock liquidity within these cyclical patterns. The paper adopts the empirical design method of Rajan and Zingales (1998), which integrates macro data to study the micro effects on firms.

(1) Cross-Market Liquidity and Market Efficiency Cyclical Two-Factor Estimation

The cyclical relationship between market liquidity and market efficiency (β_cyclical) cannot be directly determined, nor is there a direct measurement. Therefore, a first-stage regression is included to infer the cyclical changes in market liquidity and market efficiency for each month in the sample. This paper defines the estimation of market efficiency cyclicality as the change in domestic market efficiency impacted by changes in domestic market liquidity and U.S. market liquidity, reflecting the cross-market contagion to China resulting from changes in U.S. market liquidity. The model is specified as follows:

$$\triangle\ Infeff_{ch} = \alpha + \beta_{\mathrm{liqcyclical_{ch}}} \triangle\ Mktliq_{china} + \beta_{\mathrm{liqcyclical_{us}}} \triangle\ Mktliq_{us} + \varepsilon \quad (6)$$

The change in $\triangle$Mktliq reflects the position within the liquidity cycle during that period; α is the intercept term. Equation (6) estimates $\beta_{\text{liqcyclical}}$ separately for each period in the sample using rolling 2-year monthly data. A positive regression coefficient ($\beta_{\text{liqcyclical}}$) (and vice versa, negative) indicates pro-liquidity cycle market efficiency (and vice versa, counter-liquidity cycle market efficiency), meaning that during periods of declining market liquidity (positive changes in $\triangle$Mktliq), the synchronization of individual stocks with the market increases (positive changes in $\triangle$Infeff). The increase in market synchronization represents a decrease in the overall market efficiency as constructed by the average of all individual stocks in market.

Among them, $Mktliq_{china}$ represents the liquidity at the China market level, and $Mktliq_{us}$ represents the liquidity at the US market level. Liquidity is calculated using daily trading data with the monthly average of the illiquidity ratio proposed by Amihud (2002). The monthly market liquidity value is obtained by averaging the liquidity values of all sample firms for that month. Amihud (2002) measures the absolute value of returns per unit of trading volume, where a higher ratio indicates lower liquidity.

In this study, referring to the research method of Kamara, Lou, and Sadka (2008), the monthly trend in overall market efficiency ($\triangle$Infeff) is calculated in equation (2).

(2) Model of the Impact of Market Efficiency Factors Based on Market Liquidity Cycles on Stock Liquidity

In order to examine the impact of market efficiency factors based on market liquidity cycles on stock liquidity, this study regresses the stock liquidity in month t on the interaction between financial market mechanism factors and market efficiency factors from liquidity cycles. The model is specified as follows:

$$Amihud = \alpha + \beta_1 liqcyclical_{us} + \beta_2 liqcyclical_{china} + \beta_3 Z + \beta_4 liqcyclical_{us} * Z + \beta_5 liqcyclical_{china} * Z + Controls \quad (7)$$

Among them, $liqcyclical_{china}$ liqcyclical_China and $liqcyclical_{us}$ represent the market efficiency factors based on liquidity cycles, calculated from the regression coefficients $\beta_{\mathrm{cyclical_{ch}}}$ and $\beta_{\mathrm{cyclical_{us}}}$ estimated using rolling 2-year monthly data in equation (7).

Z represents the factors influencing the financial market, including: systemic liquidity (liqrsq), market interest rates.

According to the hypothesis, this study expects that β4 and β5 are significant, indicating that the market efficiency factors based on the liquidity cycle affect individual stock liquidity through financial market mechanisms.

5.2 Empirical Results

Table 5 examines the impact of market efficiency factors based on liquidity cycles in China and the United States on stock liquidity. From the empirical results in column 1, this study finds that that overall, the market efficiency factors based on liquidity cycles in both China and the United States significantly reduced liquidity risk in China. Columns 2, 3, and 4 of Table 5 test the impact of market efficiency factors based on liquidity cycles in China and the United States on stock liquidity through financial market mechanisms.

The regression results show that the market efficiency factors based on liquidity cycles, representing China and the United States, produce opposite outcomes. The China market efficiency factor based on liquidity cycles can significantly affect interest rates and liquidity risk, increasing stock liquidity. However, its impact on information risk reduces liquidity. This result reflects China's approach, which aligns with the policy objective of financial stability (Brunnermeier et al., 2022). In contrast, the effect of the U.S. market efficiency factor based on liquidity cycles is the opposite. Therefore, China needs to pay attention to the influence of U.S. market liquidity on China market efficiency.

6. Conclusion

The government faces a balancing act between ensuring financial stability and improving price efficiency. I constructed market efficiency and economic cycle factors, taking into account the global spread of financial risks from the United States. This paper constructs market efficiency and liquidity cycle factors, providing policymakers with insights into the key trade-offs faced when managing the financial system: the necessity of considering economic cycles and liquidity cycles in a timely manner when formulating corresponding policies. The micro-level financial market risk analysis in this paper emphasizes specific financial market conditions, and the empirical results reflect the advancement and scientific nature of China's financial market intervention policies, providing a reasonable explanation for the principle of prioritizing financial stability over other policy objectives.

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Table 1 Descriptive statistics

Variable	N	mean	Std	Q1	Median	Q3
Liqrsq	162250	0.2840	0.2396	0.0691	0.2333	0.4539
Cyclical_real	169701	1.9732	9.1465	-5.5542	2.1088	6.1178
Sales	168641	4.1374	7.3138	0.9417	2.1363	4.6272
Size	168622	1.4406	1.0811	0.6993	1.3484	2.0589
Ret	161282	0.0008	0.0068	-0.0032	0.0006	0.0046
Ret_std	161282	0.0276	0.0126	0.0189	0.0247	0.0330
pb	165884	4.3353	7.0879	1.9269	2.9741	4.7293
Shibor	169701	3.9493	1.2060	2.8606	3.8961	4.9749
SDE	163333	0.1818	0.2180	0.0705	0.1299	0.2265
TRD	162250	0.2555	0.2300	0.0517	0.1935	0.4165
State	169500	0.4746	0.4994	0	0	1

Table 2 The Impact of Market Efficiency, Counter-Cyclical Factors, and Systemic Liquidity on Securities Liquidity

	Liqrsq	Liqrsq	Amihud（***10**^{9}）	Amihud（***10**^{9}）
cyclical_real	0.0019***		0.0008***	
	(0.0001)		(0.0003)	
Sales	0.0023***	0.0025***	-0.0017***	-0.0012***
	(0.0002)	(0.0002)	(0.0004)	(0.0004)
Size	-0.0649***	-0.0639***	-0.3268***	-0.3290***
	(0.0016)	(0.0016)	(0.0060)	(0.0061)
Ret	-7.1294***	-7.0023***	-7.3514***	-7.1087***
	(0.0987)	(0.0978)	(0.3845)	(0.3845)
Ret_std	2.1909***	2.2354***	2.1998***	2.2591***
	(0.0669)	(0.0666)	(0.2205)	(0.2206)
pb	-0.0019***	-0.0018***	-0.0020*	-0.0020*
	(0.0002)	(0.0002)	(0.0011)	(0.0011)
cyclical_Nominal		0.0037***		0.0001
		(0.000100)		(0.0004)
Liqrsq			0.2457***	0.2241***
			(0.007300)	(0.007100)
cyclical_real*Liqrsq			0.0094***	
			(0.000700)	
cyclical_Nominal*Liqrsq				0.0182***
				(0.001100)
Intercept	0.3368***	0.3357***	0.6906***	0.6936***
	(0.0032)	(0.0031)	(0.0115)	(0.0115)
N	156750	156750	156108	156108
R-sq	0.17	0.174	0.43	0.432

The numbers in parentheses are standard errors, taking heteroscedasticity into account; *** indicates significance at the 0.01 level, ** indicates significance at the 0.05 level, and * indicates significance at the 0.1 level.

Table 3 Factors Affecting Liquidity and the Mechanism of Counter-Cyclical Factors on Market Efficiency

	Amihud （$*10^9$）	Amihud （$*10^9$）	Amihud （$*10^9$）	Amihud （$*10^9$）
cyclical_real	0.0158***	-0.0152***	0.0055***	0.0037***
	(0.0007)	(0.0007)	(0.0003)	(0.0003)
Sales	-0.0001	-0.0022***	-0.0004	-0.0012***
	(0.0004)	(0.0004)	(0.0004)	(0.0004)
Size	-0.3446***	-0.3350***	-0.3743***	-0.3429***
	(0.0061)	(0.0060)	(0.0063)	(0.0061)
Ret	-8.3819***	-8.4966***	-9.5662***	-9.1724***
	(0.3927)	(0.3781)	(0.3757)	(0.3841)
Ret_std	2.1522***	2.5502***	2.6450***	2.7690***
	(0.2012)	(0.2242)	(0.2191)	(0.2199)
pb	-0.0026**	-0.0021**	-0.0019*	-0.0024**
	(0.0011)	(0.0011)	(0.0011)	(0.0011)
cyclical_real*Ret_std	-0.4582***			
	(0.024000)			
cyclical_real*shibor		0.0049***		
		(0.0002)		
shibor		0.0623***		
		(0.001700)		
cyclical_real*SDE			-0.0090***	
			(0.0013)	
SDE			-0.0153	
			(0.019000)	
cyclical_real*TRD				0.0018***
				(0.000600)
TRD				0.0368***
				(0.006000)
Intercept	0.7808***	0.5564***	0.8445***	0.7630***
	(0.010900)	(0.012000)	(0.012000)	(0.011200)

N	156108	156108	151335	156108
R-sq	0.426	0.432	0.437	0.423

The numbers in parentheses are standard errors, taking heteroscedasticity into account; *** indicates significance at the 0.01 level, ** indicates significance at the 0.05 level, and * indicates significance at the 0.1 level.

Table 4 **Government Guarantee Effect: Transmission Channels of Market Efficiency Counter-Cyclical Factors on Liquidity Dry-Up between state-owned firms and non-state-owned firms**

	None- State own Amihud （$*10^9$）	None- State own Amihud （$*10^9$）	None- State own Amihud （$*10^9$）	None- State own Amihud （$*10^9$）	State own Amihud （$*10^9$）	State own Amihud （$*10^9$）	State own Amihud （$*10^9$）	State own Amihud （$*10^9$）
cyclical_real	0.0196***	-0.0156***	0.0055***	0.0035***	0.0120***	-0.0145***	0.0057***	0.0038***
	(0.0010)	(0.0011)	(0.0005)	(0.0004)	(0.0008)	(0.0009)	(0.0004)	(0.0004)
cyclical_real*Ret_std	-0.5943***				-0.3198***			
	(0.0374)				(0.0300)			
Ret_std	2.8813***	3.1507***	3.1935***	3.4377***	1.4956***	1.9425***	2.1221***	2.0892***
	(0.3215)	(0.3564)	(0.3512)	(0.3488)	(0.2355)	(0.2610)	(0.2574)	(0.2557)
Sales	-0.0020***	-0.0048***	-0.0020**	-0.0036***	0.0009*	-0.0008	0.0003	0.0002
	(0.0008)	(0.0009)	(0.0008)	(0.0009)	(0.0005)	(0.0005)	(0.0005)	(0.0005)
Size	-0.3622***	-0.3531***	-0.3958***	-0.3605***	-0.3207***	-0.3092***	-0.3412***	-0.3194***
	(0.0084)	(0.0084)	(0.0088)	(0.0084)	(0.0092)	(0.0091)	(0.0094)	(0.0093)
Ret	-7.2291***	-7.7283***	-8.8317***	-8.3086***	-9.9107***	-9.6387***	-10.6243***	-10.4155***
	(0.5966)	(0.5729)	(0.5787)	(0.5851)	(0.4701)	(0.4523)	(0.4461)	(0.4558)
pb	-0.0027*	-0.0022	-0.0019	-0.0025	-0.0039***	-0.0035***	-0.0035***	-0.0038***
	(0.0016)	(0.0016)	(0.0016)	(0.0016)	(0.0010)	(0.0009)	(0.0009)	(0.0010)
cylical_real*shibor		0.0051***				0.0047***		
		(0.0003)				(0.0002)		
shibor		0.0609***				0.0644***		
		(0.0027)				(0.0022)		
cylical_real*SDE			-0.0091***				-0.0096***	
			(0.0021)				(0.0015)	
SDE			-0.0470*				0.027	
			(0.0263)				(0.0224)	
cylical_real*TRD				0.0026***				0.001
				(0.0009)				(0.0007)
TRD				0.0331***				0.0403***
				(0.0096)				(0.0075)
Intercept	0.6989***	0.4876***	0.7917***	0.6878***	0.8333***	0.5964***	0.8605***	0.8125***
	(0.0157)	(0.0175)	(0.0173)	(0.0163)	(0.0165)	(0.0177)	(0.0174)	(0.0168)
N	80904	80904	76907	80904	75146	75146	74370	75146
R-sq	0.413	0.417	0.43	0.409	0.447	0.457	0.451	0.446

The numbers in parentheses are standard errors, taking heteroscedasticity into account; *** indicates significance at the 0.01 level, ** indicates significance at the 0.05 level, and * indicates significance at the 0.1 level.

Table 5 **The Impact Mechanism of Liquidity Risk and Liquidity Cycles on Market Efficiency Factors**

	Liqrsq	Amihud（*10^9）	Amihud（*10^9）	Amihud（*10^9）
Liqcyclical_china	-0.0332***	-0.0091***	-0.0479***	-0.0518***
	(0.0008)	(0.0027)	(0.0045)	(0.0026)
Liqcyclical_us	-0.0198***	0.0244***	-0.0264***	0.0257***
	(0.0007)	(0.0018)	(0.0025)	(0.0017)
Sales	0.0022***	-0.0019***	-0.0025***	-0.0006
	(0.0002)	(0.0004)	(0.0004)	(0.0004)
Size	-0.0688***	-0.3246***	-0.3335***	-0.3708***
	(0.0016)	(0.0060)	(0.0060)	(0.0063)
Ret	-6.5930***	-6.9981***	-7.9746***	-9.2514***
	(0.0977)	(0.3858)	(0.3809)	(0.3787)
Ret_std	1.2963***	2.5904***	2.2522***	2.7906***
	(0.0593)	(0.2296)	(0.2345)	(0.2222)
pb	-0.0020***	-0.0020*	-0.0022**	-0.0019*
	(0.0002)	(0.0010)	(0.0011)	(0.0011)
Liqrsq		0.4590***		
		(0.0153)		
Liqrsq*Liqcyclical_us		0.0067**		
		(0.0028)		
Liqrsq*Liqcyclical_china		-0.0619***		
		(0.0046)		
shibor*Liqcyclical_us			0.0149***	
			(0.0007)	
shibor*Liqcyclical_china			-0.0052***	
			(0.0011)	
shibor			0.1130***	
			(0.0037)	
SDE*Liqcyclical_us				-0.0095***
				(0.0029)
SDE*Liqcyclical_china				0.0807***

				(0.0088)
SDE				-0.2684***
				(0.0316)
Intercept	0.4762***	0.6337***	0.6372***	0.9333***
	(0.0042)	(0.0147)	(0.0180)	(0.0149)
N	156750	156108	156108	151335
R-sq	0.183	0.434	0.436	0.441

The numbers in parentheses are standard errors, taking heteroscedasticity into account; *** indicates significance at the 0.01 level, ** indicates significance at the 0.05 level, and * indicates significance at the 0.1 level.